



Fiona Stanley Fremantle Hospitals Group

Guideline

Smokerlyzer®: Therapeutic use

Reference #: FSFH-MENH-GUI-0010

Scope

Site	Service/Department/Unit	Disciplines
Fiona Stanley Hospital Fremantle Hospital	Mental Health Service	Medical, Nursing, Allied Health

1. Introduction

- A Smokerlyzer® can be used by mental health clinicians as a motivational tool to support and enhance the delivery of smoking cessation advice to inpatients. The purpose of this guideline is to inform clinicians how to use and maintain the Smokerlyzer®.

2. Terminology

Carbon Monoxide (CO)	An odourless, colourless, poisonous gas produced by burning organic material. CO in addition to tar and nicotine are the main constituents of tobacco smoke and represent a risk to health. When smoke is inhaled from a cigarette CO is absorbed into the blood through the lungs. The more a patient smokes the higher their CO reading. Once a patient stops smoking their CO levels drop quickly and within 24-48 hours CO levels go back to the level of a non-smoker.
Smokerlyzer®	 Handheld breath CO device intended for multi-patient use by healthcare professionals in smoking cessation programmes. The brand used at FSFHG is a Smokerlyzer® piCO™.

Breath sampling D piece	 Bacterial filter used to attach a single use mouthpiece to the device. Contains a one-way valve to prevent patients drawing air back from the device. The integrated infection control filter removes and traps > 99% of airborne bacteria and >96% of viruses. The D piece should be replaced every 28 days or if visibly soiled.
Mouthpiece	Single use mouthpiece that connects directly to the D piece.
Carboxyhemoglobin (COHb)	A stable complex of carbon monoxide that forms in red blood cells when carbon monoxide is inhaled. Non-smokers < 3%

3. Smokerlyzer®

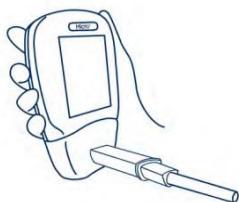
3.1. Indications for use

- A Smokerlyzer® reading may be taken to:
 - screen for nicotine dependence on admission
 - confirm smoking status
 - establish initial expired CO level
 - measure CO levels after a patient smokes whilst off the ward to demonstrate the immediate effects of smoking.
 - compare against readings after a period of abstinence later in the admission.

Do not use the Smokerlyzer® on patients with symptoms of influenza like illness and/or COVID19

3.2. Instructions for use

- Obtain patient consent
- Clinician and patient to wash hands before use
- Turn on the monitor by pressing the power button once
- Press 'breath test' symbol on the screen
- Attach a breath sampling D-piece with a new mouthpiece



- Inhale and hold breath or the pre-set 15 second countdown
- Press the home button  at any time to cancel the breath test
- A beep will sound during the last 3 seconds of the countdown
- Ask the patient to blow slowly into the mouthpiece, aiming to empty their lungs completely
- The ppm and equivalent % COhb and/or %fCOhb levels will rise and hold onscreen
- When the test is finished   will appear at the bottom of the screen
- If a high reading has been recorded press  to mute the sounder
- To repeat the breath test, press  once to return to home screen and repeat steps
- Remove the D piece between tests to purge the sensor with fresh air
- To switch off press and hold the power button for 3 seconds, unit will also power off after 2 minutes of inactivity to save power.
- Clinician and patient to wash hands after use.
- Document all Smokerlyzer® readings in the patients integrated progress notes and in their Quitline 'Smoking Diary' page 24-25.

3.3. Interpreting Smokerlyzer® readings

Traffic light	COppm	What does this mean
Green	0 - 6	• Non smoker
Amber	7 - 9	• Borderline
1 Red	10 - 15	• Smoker – low addicted
2 Red	16 - 25	• Smoker - moderately addicted
3 Red	26 - 35	• Smoker – heavily addicted
3 Red flashing	36+	• Smoker – very heavily addicted

- The more CO in the patient's breath and bloodstream, the less space there is for oxygen.
- In addition to leading to diseases of the lungs like cancer and emphysema, it also means the heart must work harder to send oxygen around the body and increases the likelihood of myocardial infarction.
- Inform the patient that it is a positive indication when their CO levels start to go down as it means their lungs, bloodstream and organs are exposed to less poisonous gas, and more oxygen can get around their body.
- Appropriate interventions may include:
 - Brief advice
 - Nicotine replacement therapy
 - Behavioural supports
 - Referral to appropriate to Alcohol and Drug Service and / or Quitline

3.4. Cleaning the Smokerlyzer®

- The Smokerlyzer® should be cleaned prior to initial use and after each patient use
- Clean the device with hospital approved wipes

Never use alcohol or cleaning products containing alcohol or other organic solvents as these vapours will damage the electrochemical sensor within the device.

4. Compliance/performance monitoring

- The Nurse Unit Manager/Clinical Nurse Specialist of the area is responsible for ensuring the Smokerlyzer® is used and maintained in compliance with this guideline.

5. Related policy documents

WA Health:

- [Smoke free Policy MP 0158/21](#)
- [Guidelines for the implementation of the smoke free policy](#)

FSFHG:

- Management of nicotine dependence in inpatients (FSFH-HW-GUI-0026)
- Environmental Cleaning (FSFH-HW-POL-0002)

6. Related documents

- [LAB679 Smokerlyzer-manual-issue 13](#)

7. Related standards

NSQHS Standards:

- Clinical Governance
- Partnering with consumers
- Preventing and controlling healthcare-associated infection
- Comprehensive care

Chief Psychiatrists Standards for Clinical Care:

- Physical health

8. Authorisation

EXECUTIVE SPONSOR: Service Director, Service 5					
Version	Date Issued	Compiled/Revised By	Committee/Consumer Group Consulted	Authorised By	Revision due
1	01/2021	CNS, CI	S 5 Safety Quality Risk Committee	SQR Committee, Service 5	01/2024
2	02/2024	CNS, CI		SQR Committee, Service 5	02/2028